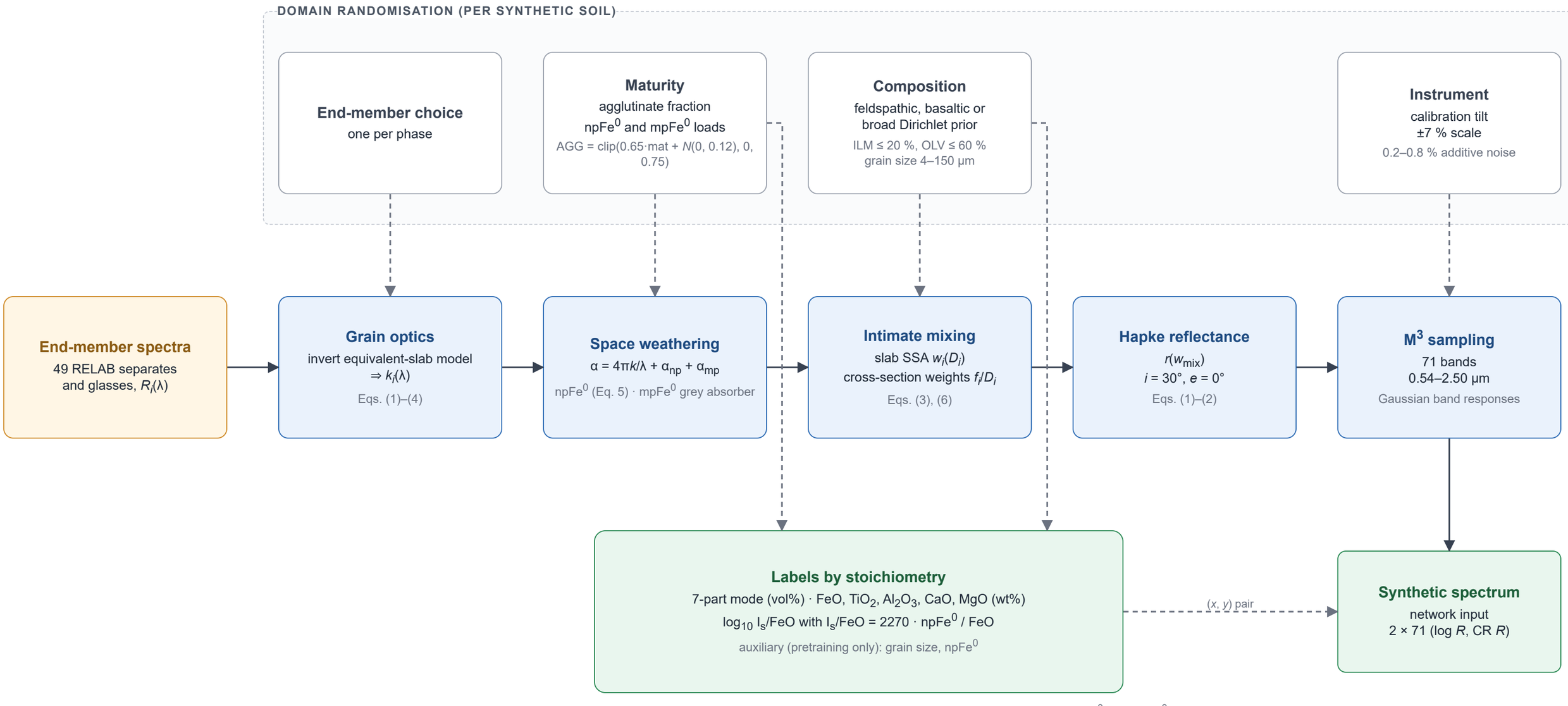

DOMAIN RANDOMISATION (PER SYNTHETIC SOIL)
End-member choice
one per phase
Maturity
agglutinate fraction
npFe0 and mpFe0 loads
AGG = clip(0.65·mat + N(0, 0.12), 0, 0.75)
Composition
feldspathic, basaltic or broad Dirichlet prior
ILM ≤ 20 %, OLV ≤ 60 %
grain size 4–150 µm
Instrument
calibration tilt
±7 % scale
0.2–0.8 % additive noise
End-member spectra
49 RELAB separates and glasses, Ri(λ)
Grain optics
invert equivalent-slab model
⇒ ki(λ)
Eqs. (1)–(4)
Space weathering
α = 4πk/λ + αnp + αmp
npFe0 (Eq. 5) · mpFe0 grey absorber
Intimate mixing
slab SSA wi(Di)
cross-section weights fi/Di
Eqs. (3), (6)
Hapke reflectance
r(wmix)
i = 30°, e = 0°
Eqs. (1)–(2)
M3 sampling
71 bands
0.54–2.50 µm
Gaussian band responses
Labels by stoichiometry
7-part mode (vol%) · FeO, TiO2, Al2O3, CaO, MgO (wt%)
log10 Is/FeO with Is/FeO = 2270 · npFe0 / FeO
auxiliary (pretraining only): grain size, npFe0
(x, y) pair
Synthetic spectrum
network input
2 × 71 (log R, CR R)
400,000 spectra, fixed seed · no end-member from an evaluation soil · agglutinates are a synthetic-glass host carrying npFe0 and mpFe0